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3 CONTROLS AND INDICATORS
3.1 Subject messages2 excluded by the operator from display should be readily available.
3.2 When a message is received correctly or with errors, the equipment may be able to emit a corresponding short audio notification. In addition, a distinctive alert should be built into the receiver for messages with distress and urgency priority.
3.3 The NAVDAT receiver equipment should be capable of displaying communication parameters, including received signal strength indication (RSSI), signal noise ratio (SNR) and bit error rate (BER).
3.4 Each time a NAVDAT message is received, the NAVDAT receiver equipment should display, in plain text, the priority, subject message, coast station identity and number of the message.
3.5 The NAVDAT receiver equipment should be able to configure parameters of data interfaces for communication with other ship equipment.
3.6 The stored tables of the NAVDAT receiver equipment should be able to be updated
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interfaces for communication with other ship equipment.
3.6 The stored tables of the NAVDAT receiver equipment should be able to be updated by using either a data interface or reception of stored tables update messages.
3.7 Human-machine interface, including the presentation of alerts, for the NAVDAT receiver equipment should be in accordance with the guidelines developed by the Organization.3
4 RECEIVER
4.1 Frequency management
4.1.1 The NAVDAT receiver should be capable of receiving transmissions on MF (500 kHz) and HF (4 226 kHz) channels simultaneously.
4.1.2 The NAVDAT receiver should also be able to receive, via a scanning function, on at least one (or more) regional frequency allocated to NAVDAT in the MF and HF maritime bands.
4.1.3 The NAVDAT receiver should demodulate signal(s) received on the MF channel 500 kHz and the HF channel 4 226 kHz simultaneously.
4.1.4 The demodulated signal(s) received by scanning can be decoded simultaneously or
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500 kHz and the HF channel 4 226 kHz simultaneously.
4.1.4 The demodulated signal(s) received by scanning can be decoded simultaneously or non-simultaneously.
4.2 Scan function
4.2.1 To allow reception of transmissions on national or regional frequencies assigned to NAVDAT, the receiver should use a scan function on the following maritime frequency bands:
.1 the MF band from 415 kHz to 526.5 kHz (except 500 kHz);
2 See the list of subject message codes in the most recent version of Recommendation ITU -R M.2010 or ITU-R M.2058.
3 Refer to Guideline on software quality assurance and human -centred design for e-navigation

(MSC.1/Circ.1512).
RESOLUTION MSC.569(109) (adopted on 6 December 2024)
PERFORMANCE STANDARDS FOR THE RECEPTION OF MARITIME SAFETY
INFORMATION AND SEARCH AND RESCUE RELATED INFORMATION
BY MF AND HF DIGITAL NAVIGATIONAL
DATA (NAVDAT) SYSTEM
MSC 109/22/Add.1
Annex 12, page 4
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.2 the channels assigned to NAVDAT in appendix 15 of the Radio Regulations:
6 337.5 kHz, 8 443 kHz, 12 663.5 kHz, 16 909.5 kHz and 22 450.5 kHz
(except 4 226 kHz); and
.3 the frequency bands assigned to wideband digital transmissions in
appendix 17 of the Radio Regulations in the bands 4, 6, 8, 12, 16
and 22 MHz.
4.2.2 The signals received on the frequencies selected for scanning can be decoded
simultaneously or non-simultaneously according to the resources of the NAVDAT receiver at
that moment.
4.2.3 The transmitted pre -signal4 should allow the NAVDAT receiver to detect the
transmission and tune in to the frequency, measure its SNR and identify the transmitting
NAVDAT coast station, including, based on its location, the associated NAVAREA/METAREA.
4.3 The receiver sensitivity should be better than 95dBm in a bandwidth of 10 kHz, with
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4.3 The receiver sensitivity should be better than 95dBm in a bandwidth of 10 kHz, with
a BER better than 10-4 after error correction.
4.4 The NAVDAT receiver equipment should comply with the latest appropriate
electromagnetic compatibility (EMC) standards (e.g. resolutions A.694(17) and A.813(19), and
IEC 60945).
4.5 The NAVDAT receiver equipment should be able to receive any NAVDAT message
and identify the priority level and the type of that message.
4.6 For identification purposes, the NAVDAT receiver equipment should use the Maritime
Mobile Service Identity (MMSI) of the ship (see also paragraph 11).
4.7 The NAVDAT receiver equipment should be able to select automatically whether to
receive subsequent messages according to the following broadcast modes:
.1 General broadcast: Broadcasting messages to all ships.
.2 Selective broadcast: Broadcasting messages to a group of ships or ships
within a specific navigational area.

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.2 Selective broadcast: Broadcasting messages to a group of ships or ships
within a specific navigational area.
.3 Dedicated broadcast: Addressing messages to one ship, using MMSI.
4.8 The NAVDAT receiver equipment should be able to select automatically demodulation
mode, error correction decoding and message decoding methods according to the received
modulation information stream (MIS).
4.9 NAVDAT message transmission is based on the transmission of data packets.
The equipment should be able to correct received messages using forward error correction
and by using repeatedly sent data packets.
4.10 The NAVDAT receiver equipment should have a built -in real -time clock that is
automatically calibrated by a suitable electronic position-fixing aid (or clock reference through
an onboard data network).
4 To ensure proper operation of the receiver scan function, the transmitters of active national or regional
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an onboard data network).
4 To ensure proper operation of the receiver scan function, the transmitters of active national or regional
NAVDAT coast stations will broadcast a pre -signal before the NAVDAT frames. This pre -signal is a known
data repeated eight times for a total duration of 3.2 seconds.
RESOLUTION MSC.569(109) (adopted on 6 December 2024)
PERFORMANCE STANDARDS FOR THE RECEPTION OF MARITIME SAFETY
INFORMATION AND SEARCH AND RESCUE RELATED INFORMATION
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5 DISPLAY DEVICE
5.1 The display device should be able to display different types of NAVDAT messages.
5.2 The design and size of the integrated display device should be such that displayed
information is easily read under all conditions by observers at normal working distance and
viewing angles. An interface to an external display device or appropriate navigat ional
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viewing angles. An interface to an external display device or appropriate navigat ional
equipment should be provided.
5.3 On the display, the following requirements should be met:
.1 an indication of newly received and unsuppressed messages should be
immediately displayed until acknowledged or for 24 hours after their receipt;
and
.2 newly received unsuppressed messages should also be displayed.
5.4 When displaying received messages, a clear indication of the end of any message
should be given.
5.5 The display resolution should be at least 640*480 pixels.
5.6 The list of messages received on 500 kHz and 4 226 kHz can be displayed in different
areas of the display screen at the same time.

5.7 In the list of received messages, the latest message is displayed on the first line with its number, contents, date and time.

6 STORAGE

6.1 Non-volatile memory

6.1.1 The NAVDAT receiver equipment should keep all information concerning NAVDAT

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6 STORAGE

6.1 Non-volatile memory

6.1.1 The NAVDAT receiver equipment should keep all information concerning NAVDAT

coast stations in a non-volatile memory. This information should include name of the stations, their geographical positions, their numbering allocated by the NAVDAT coordination procedure, the frequency(s) used as well as the different slots that can be used by the stations. This information will be regularly updated upon receipt of stored tables update messages.

6.1.2 For each frequency provided, it should be possible to record at least 100 messages in the non-volatile memory provided. It should not be possible for the user to erase messages from the memory. When the memory is full, the oldest message should be replace d by new messages.

6.1.3 The user should be able to mark the individual messages for permanent retention. These messages can occupy up to 25% of available memory and should not be overwritten by

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These messages can occupy up to 25% of available memory and should not be overwritten by new messages. When no longer needed, the user should be able to delete the tag on the se messages, which can be overwritten normally.

6.1.4 The NAVDAT receiver equipment should be able to detect duplicate messages and discard repeated messages.

6.1.5 A memory capacity of not less than 1 gigabyte should be provided to store, as a minimum, the time, transmitter identity, subject and content of received messages.